Customer Segmentation Marketing Campaign Analysis

Unsupervised machine learning with Python

Project documentation and findings

2,212	23	3	4
customers analysed	features used for clustering	PCA components	customer clusters

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October 2026

1. Project Overview

This project uses unsupervised machine learning in Python to split 2,212 supermarket customers into 4 groups (clusters) based on who they are and how they spend, so that each group can receive a different marketing strategy.

The problem. Sending the same offer to every customer wastes money. A high-income couple with no children and a low-income family with three children want very different things.

The objective. Find natural customer groups in the data, describe each group in plain business terms, and recommend a marketing action for each.

The approach. Clean the data, create meaningful features, scale and compress it, choose the number of clusters, form the clusters, and then profile them.

The result. Four clear segments: older parents with teenagers, younger parents with small children, high-income customers without children, and lower-income families who look for savings. The biggest spenders are high-income customers without children.

Type of problem. Customer segmentation using clustering. It is unsupervised because the data has no labels saying which customer belongs to which group; the algorithm finds the pattern itself.

2. Dataset

The dataset (marketing_campaign.csv, tab-separated) has 2,240 customers and 29 columns. After cleaning, 2,212 customers remained. Customers joined between 30 July 2012 and 29 June 2014.

Column group	Examples	What it tells us
Personal	Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome	Who the customer is
Customer history	Dt_Customer, Recency	When they joined; days since last purchase
Spending	MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds	Amount spent on each product type
Purchase channel	NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth	How and where they buy
Campaigns	AcceptedCmp1 to AcceptedCmp5, Response, Complain	Whether they accepted past promotions
Constants	Z_CostContact, Z_Revenue	Same value for everyone, so removed

Only Income had missing values (24 of 2,240 rows).

3. Tool Stack

Tool	Used for
Python (Jupyter Notebook)	Writing and running all the code
pandas, NumPy	Loading, cleaning and calculating on data
Matplotlib, Seaborn	All graphs
scikit-learn	Encoding, scaling, PCA and clustering (K-Means, Agglomerative)
Yellowbrick	The elbow graph for choosing the number of clusters

Note: the library is called scikit-learn; sklearn is only its short import name in the code.

4. Methodology: What Was Done and Why

The project followed 13 steps, from raw data to cluster profiles.

#	Step	What was done	Why	Alternative
1	Load data	Read the file with a tab separator; 2,240 rows	File is tab-separated, not comma-separated	None needed
2	Handle nulls	Dropped 24 rows with missing Income; 2,216 left	Models cannot handle blanks; about 1% of rows	Fill with median income
3	Fix dates	Converted Dt_Customer to a date; created Customer_For (how long each person has been a customer)	A raw date means nothing to a model; tenure does	Use year of joining
4	Feature engineering	Created Age, Spent, Living_With, Children, Family_Size, Is_Parent; simplified Education into 3 levels	Better features give better clusters than a fancier algorithm	Keep all original columns
5	Clean columns	Renamed product columns; dropped ID, Year_Birth, Marital_Status, Dt_Customer, Z_CostContact, Z_Revenue	Removed duplicates of new features and constant columns	None
6	Remove outliers	Found Age 133 and Income about 666,000; kept Age below 90 and Income below 600,000; 2,212 left	Outliers pull cluster centres away from real groups	Cap values instead of deleting
7	Label encoding	Converted Education and Living_With to numbers	Models need numbers	One-hot or ordinal encoding
8	Drop campaign columns	Removed AcceptedCmp1-5, Complain, Response before clustering	Group by profile and spending first, then check campaign response afterwards	Keep them in
9	Scaling	StandardScaler: every column gets mean 0 and spread 1	Income (tens of thousands) would otherwise overpower Kidhome (0 to 2)	MinMaxScaler
10	PCA	Reduced 23 columns to 3 components	Less noise, more meaningful distances, and a 3D view	t-SNE, UMAP, feature selection
11	Elbow method	Ran K-Means for k = 2 to 10; curve bends at k = 4	Clustering needs the number of groups chosen first	Silhouette score
12	Final clustering	Agglomerative Clustering with 4 clusters on the 3 PCA columns	Builds groups by merging the most similar customers	K-Means, DBSCAN, Gaussian Mixture

#	Step	What was done	Why	Alternative
13	Profile clusters	Plotted sizes, income vs spending, promotions, deals and personal traits	Turn numbers into business meaning	Tables of cluster averages

Engineered features

- **Age** = 2026 minus birth year.
- **Spent** = total spent on wines, fruits, meat, fish, sweets and gold.
- **Living_With** = Partner (married or together) or Alone (single, divorced, widow, alone, absurd, YOLO).
- **Children** = Kidhome + Teenhome.
- **Family_Size** = 1 or 2 adults + Children.
- **Is_Parent** = 1 if Children is above 0.
- **Customer_For** = days between a customer's joining date and the newest joining date in the data.

Outlier detection

The describe() summary showed a maximum age of 133, which is impossible. The pairplot below confirms it: a few lone points sit far from everyone else on Age and Income. Removing them left 2,212 customers.

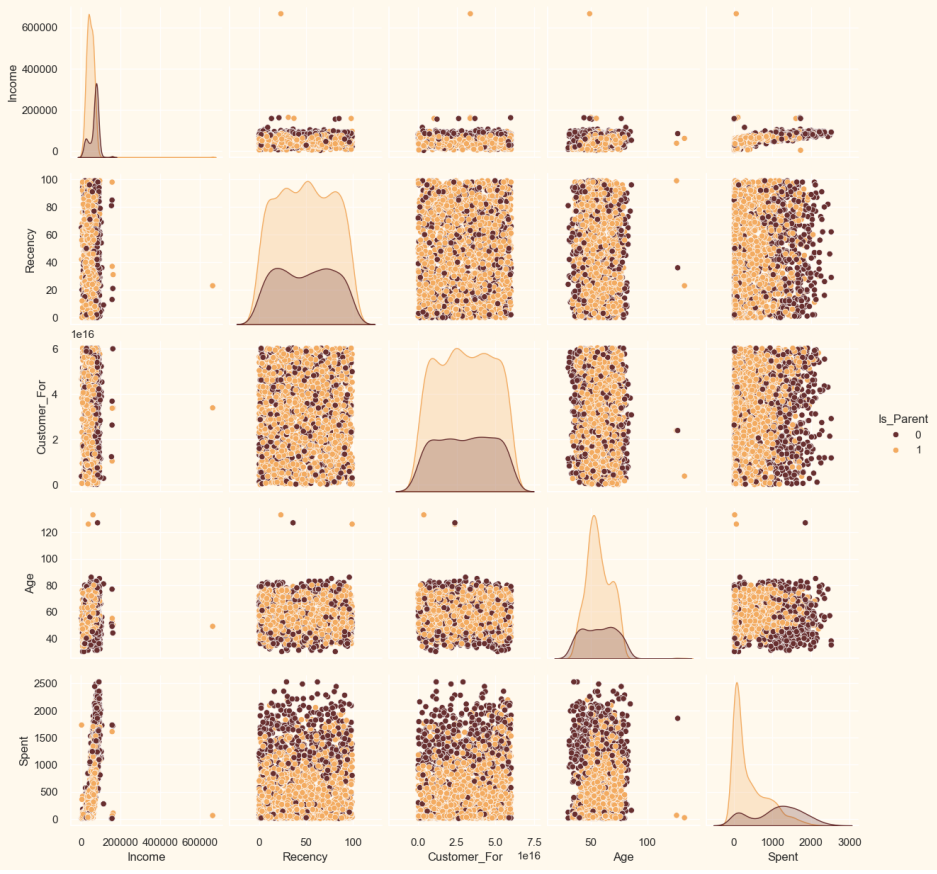


Figure 1. Pairplot of Income, Recency, Customer_For, Age and Spent, coloured by parent status. The isolated points at Age above 120 and Income near 666,000 are the outliers that were removed.

Dimensionality reduction (PCA)

Scaling put all features on the same footing, and PCA then compressed 23 columns into 3 components that keep most of the information. The 3D view below shows the whole customer base in the reduced space, already hinting at separate groups.

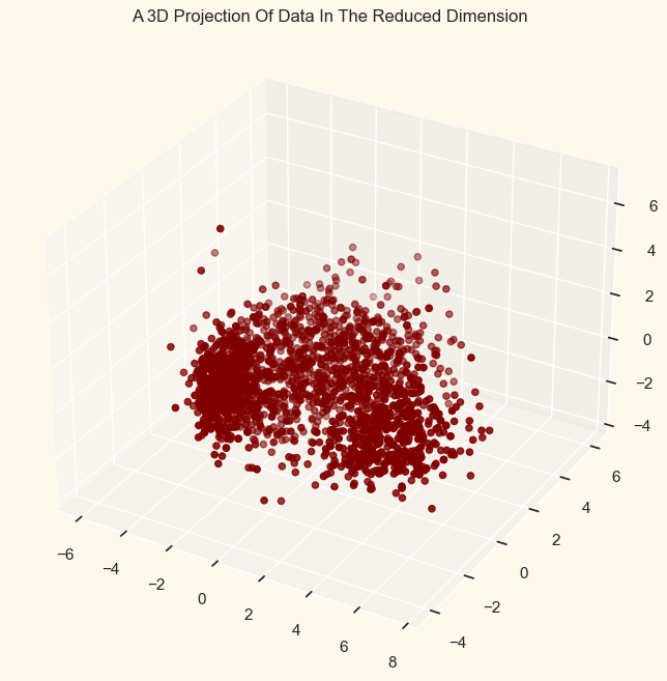


Figure 2. All customers projected onto the 3 PCA components.

Choosing the number of clusters

K-Means was run for k = 2 to 10. The distortion score falls steeply until k = 4 and only slowly after that, so 4 is the sensible number of clusters. The final groups were then formed with Agglomerative Clustering: every customer starts as their own group, and the two most similar groups merge repeatedly until 4 remain.

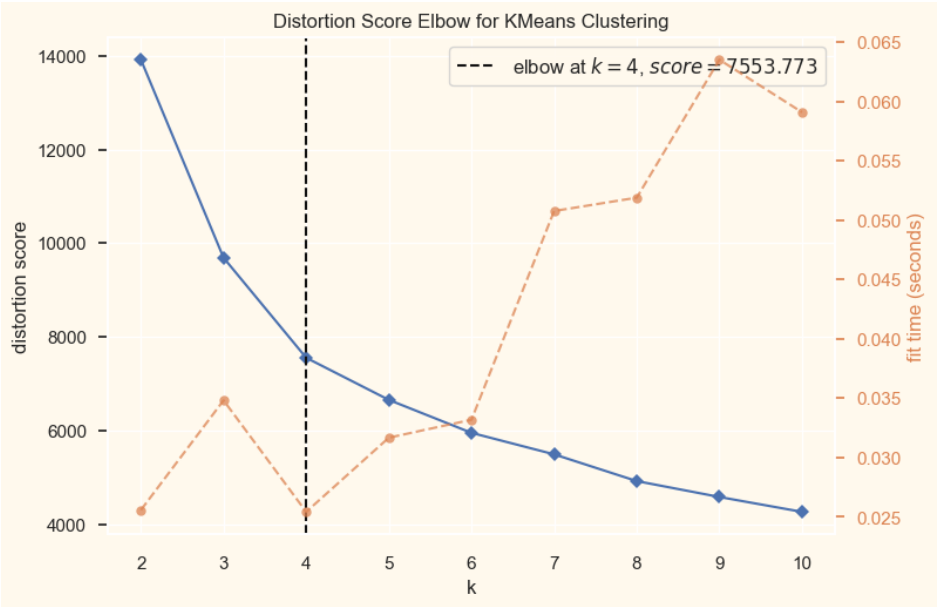


Figure 3. Elbow method: the bend at k = 4 marks the best number of clusters.

5. Results: The Four Customer Clusters

The final groups are well balanced, with roughly 500 to 600 customers each, and they separate clearly in the 3D plot and in the income versus spending plot.

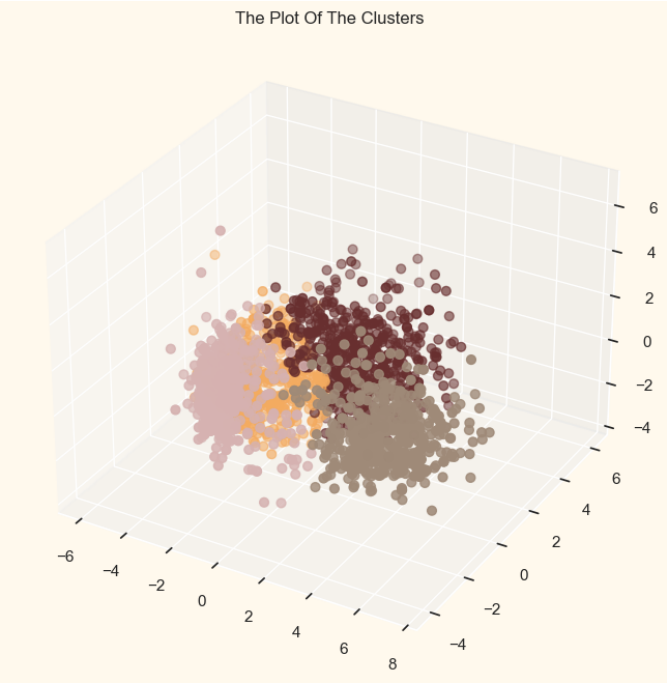


Figure 4. The four clusters in the 3 PCA dimensions.

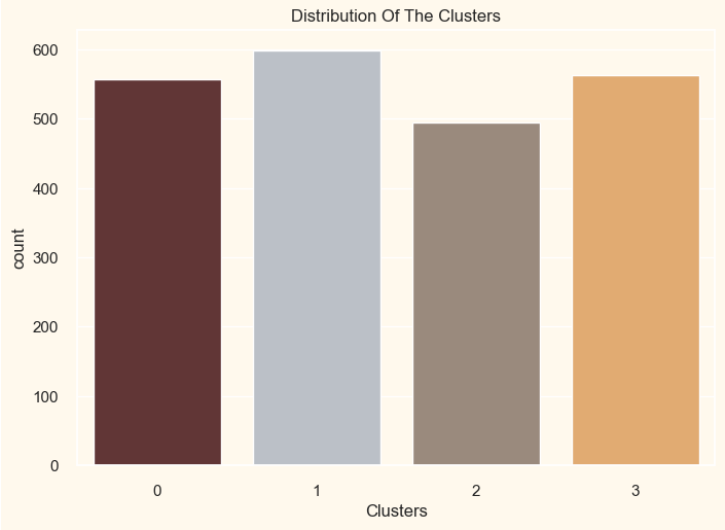


Figure 5. Number of customers in each cluster.

Spending and income

Income and spending rise together, but the clusters are separated by family situation as well as by money. Cluster 2 sits at the top for both spending and income, while Clusters 1 and 3 are concentrated at low spending.

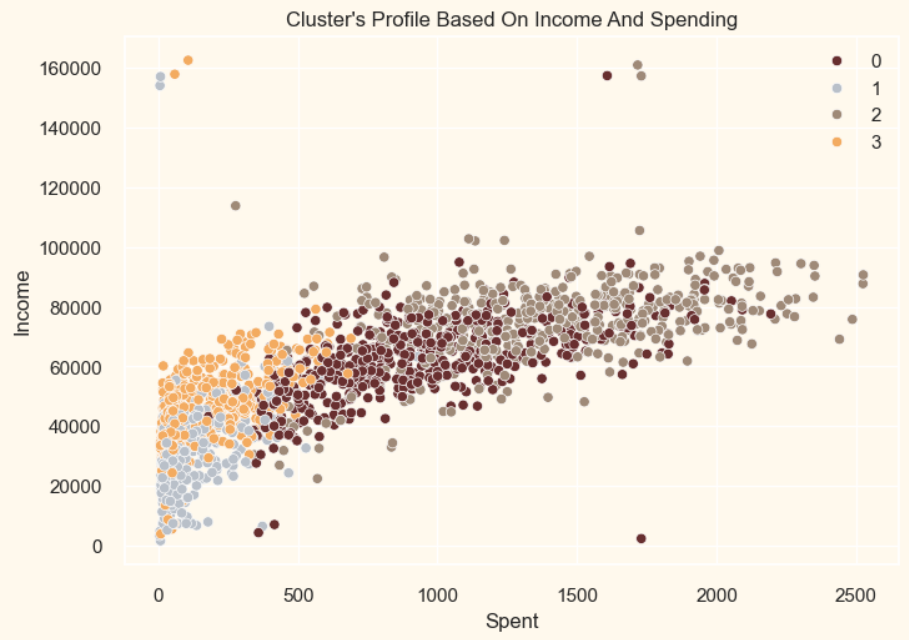


Figure 6. Income against total spending, coloured by cluster.

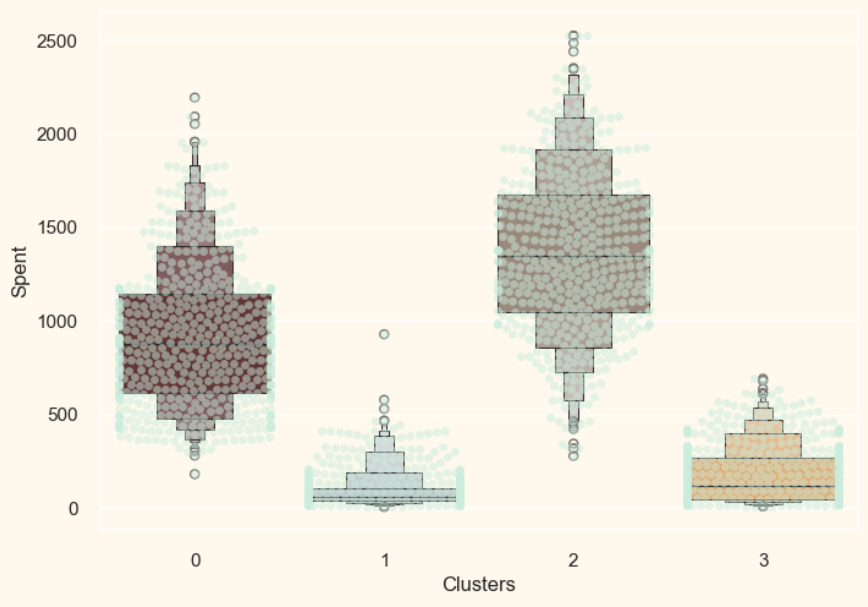


Figure 7. Spending distribution per cluster. Cluster 2 spends the most, Cluster 0 is next, Clusters 1 and 3 spend the least.

Campaign and deal behaviour

Checking campaign response after clustering shows the groups really behave differently. Cluster 2 accepted the most promotions, while Cluster 0 made the most deal purchases.

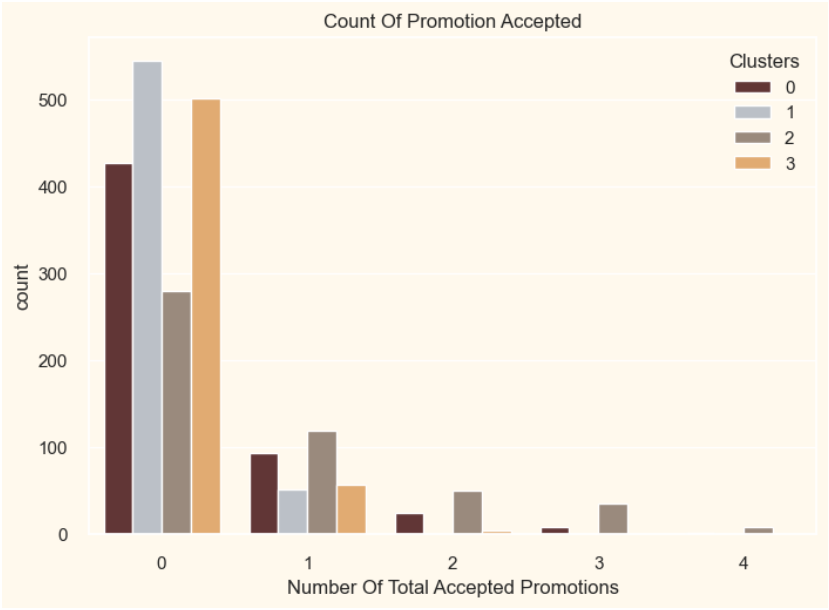


Figure 8. Total promotions accepted per cluster.

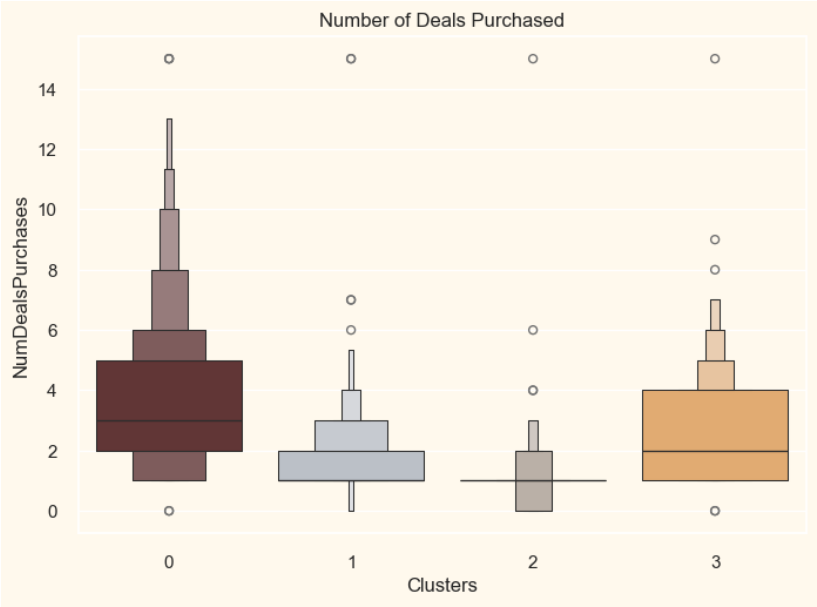


Figure 9. Number of deals purchased per cluster.

Family profile

Children, family size and parent status are what separate the groups most clearly. Cluster 2 has no children, while Clusters 0, 1 and 3 are all parents with different family sizes and spending.

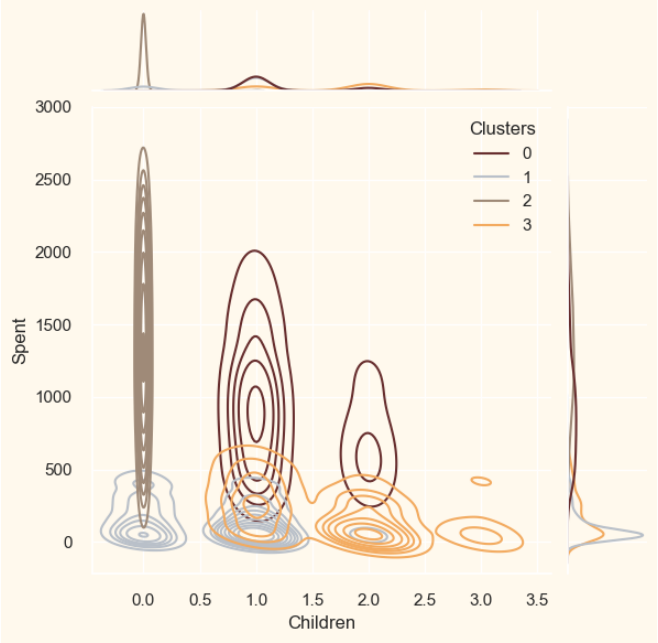


Figure 10. Spending against number of children, by cluster.

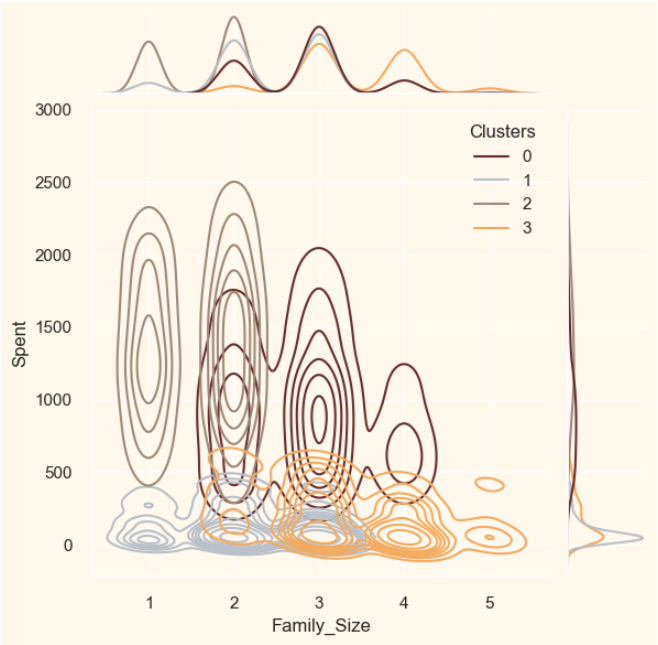


Figure 11. Spending against family size, by cluster.

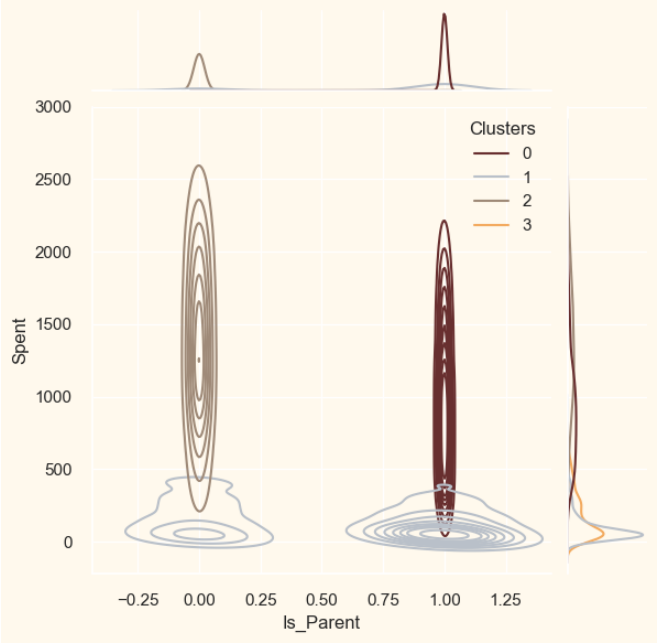


Figure 12. Spending for parents versus non-parents, by cluster.

Cluster profiles

Cluster	Who they are	Spending	Income	Behaviour seen
0	Older parents, mostly with a teenager; family of 2 to 4	Medium to high	Medium to high	Most deal purchases of any group
1	Younger parents, usually one young child; family up to 3	Lowest	Low to medium	Few promotions accepted
2	Not parents; 1 to 2 people; couples slightly more than singles; all ages	Highest	Highest	Accepts the most promotions; buys few deals
3	Older parents with teenagers; family up to 5	Low	Lower	Many deal purchases; large families

Key insights

- The biggest spenders are high-income customers without children (Cluster 2).
- Parents spend less, and those with teenagers hunt for deals the most (Clusters 0 and 3).
- Family situation separates groups that have similar income, so it matters as much as money.
- Cluster 2 responded best to promotions, so campaigns can be aimed at them rather than at everyone.
- Education and living situation showed little difference between clusters; family structure drives the groups far more.

6. Final Business Recommendations

Each cluster gets its own goal and action, based on its profile.

Cluster	Goal	Recommended actions
0 Older parents	Convenience and family value	Multi-buy offers on household staples ("Buy 2, Get 1 Free"); meal kits for 3 to 4 people; flyers and newspaper inserts alongside digital ads, since this group is older
1 Younger parents	Child-focused, quality, easy shopping	Highlight fresh produce and organic baby and toddler items; personalised app coupons; social media campaigns; kids-eat-free samples and family parking
2 High-income, no children	Premium and high margin	Promote high-end meat, imported cheese, fine wines and specialty foods; premium bundles ("Gourmet Pasta Night"); elegant email newsletters; stress quality rather than discounts
3 Lower-income families	Savings and volume	Private-label products; bulk and family-size packs; weekly circular sales; loyalty points or instant cash-off rewards